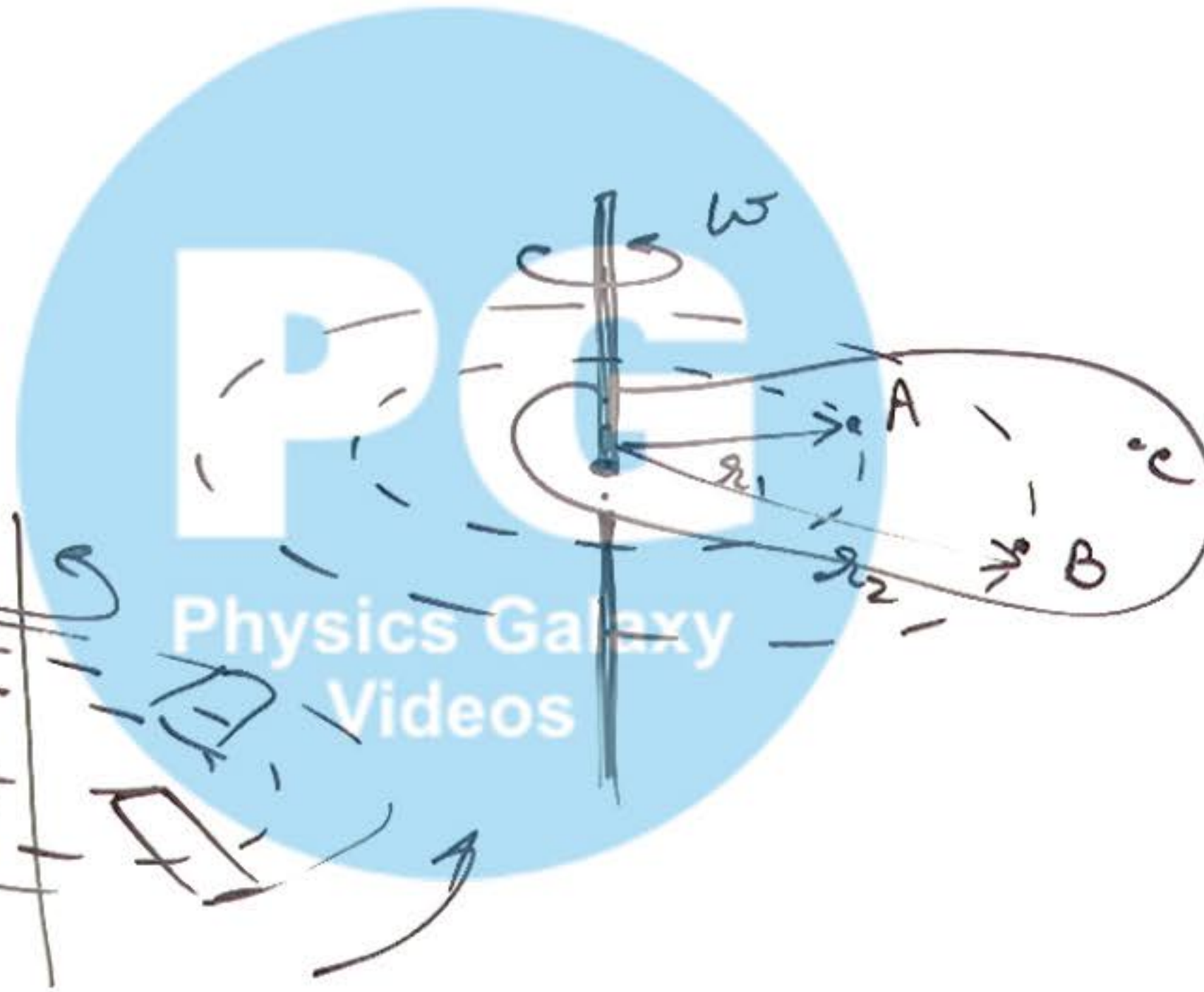


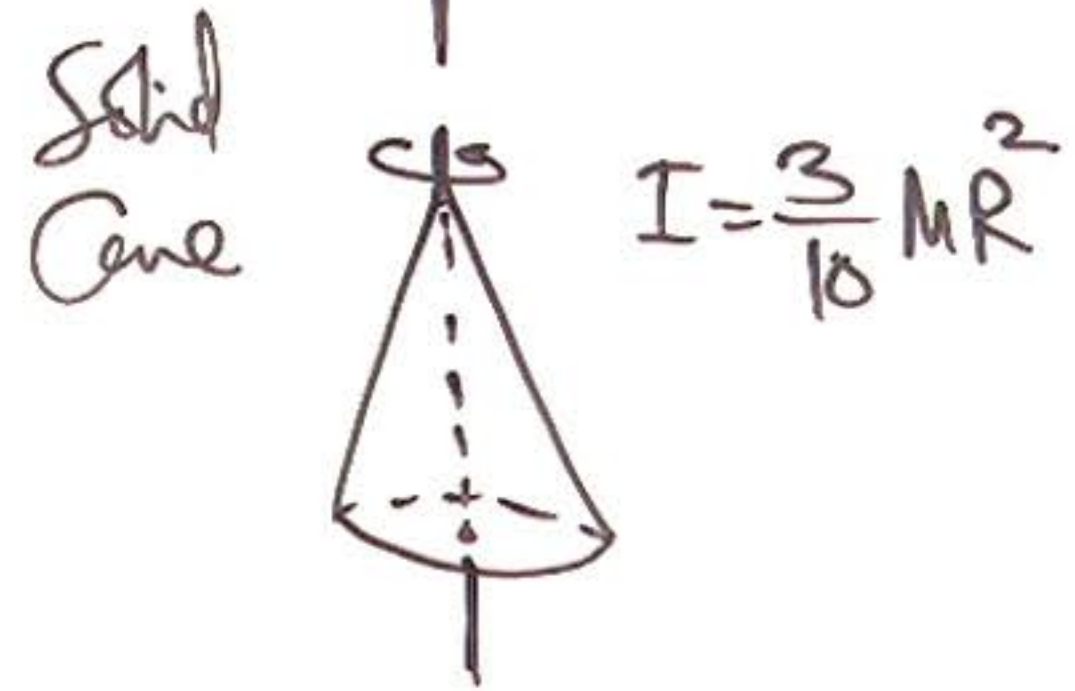
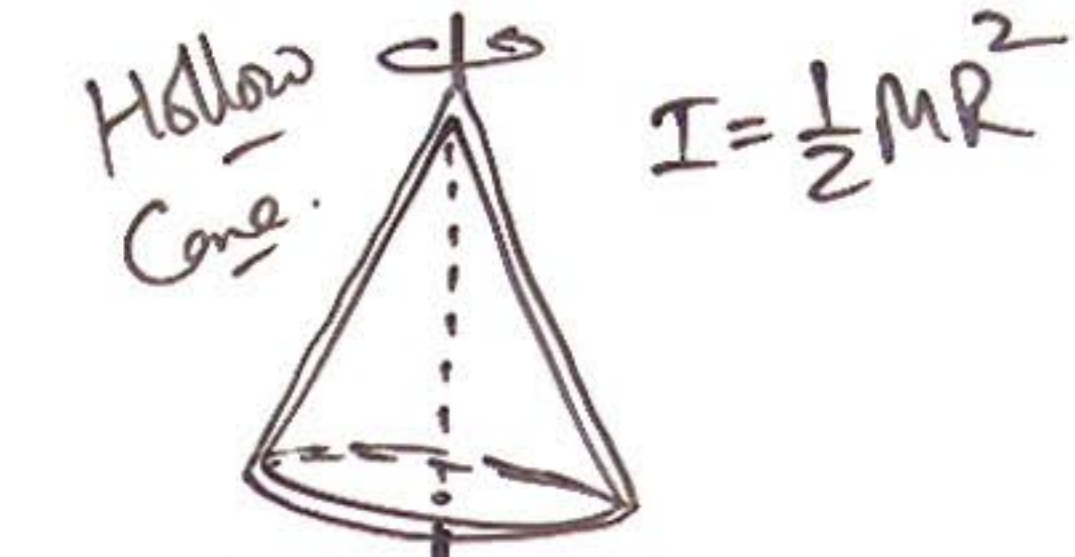
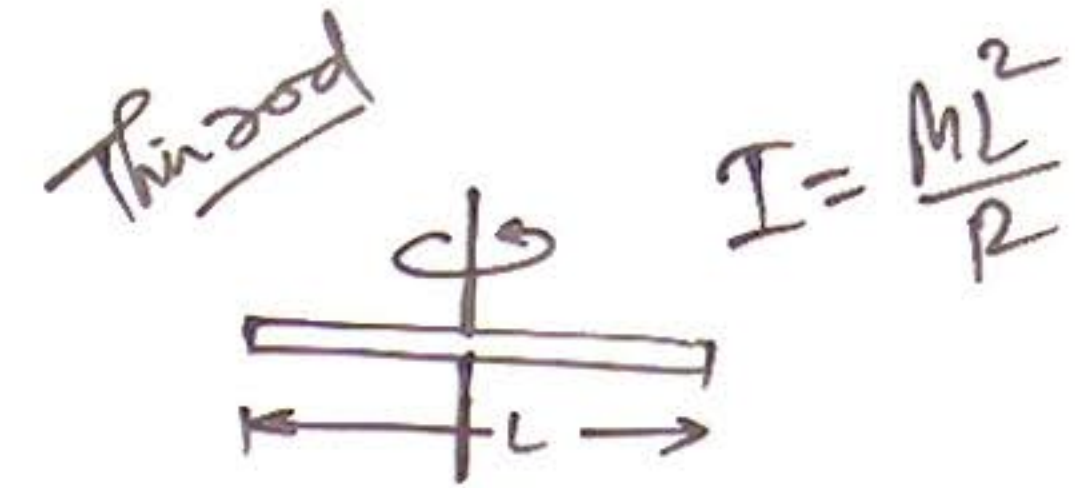
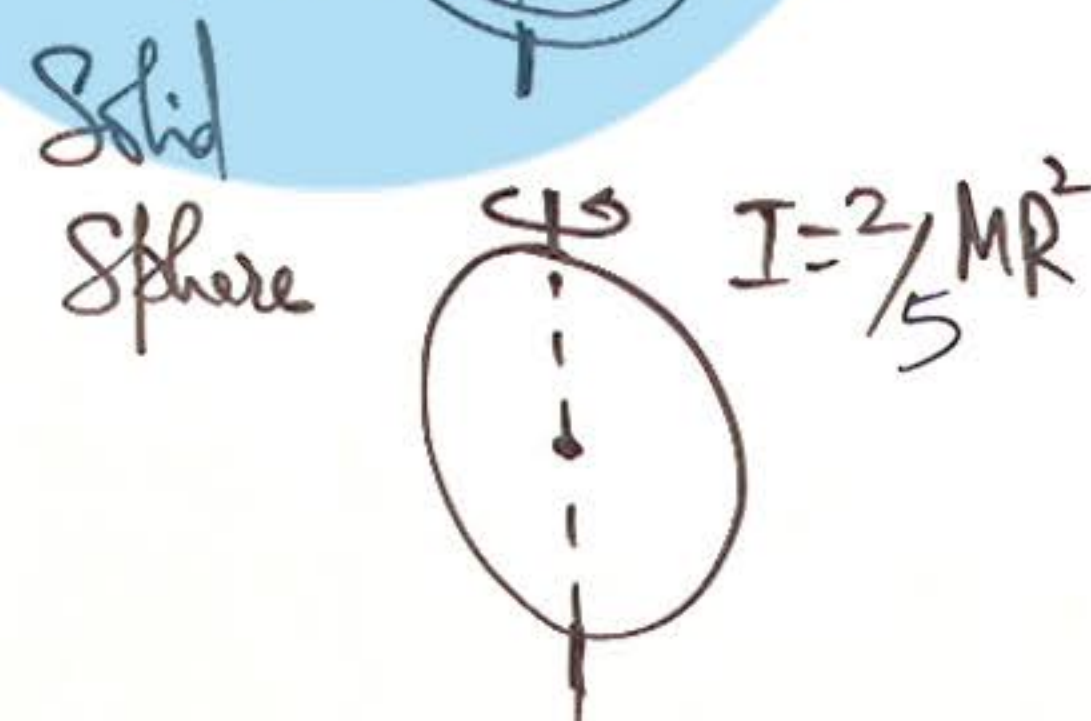
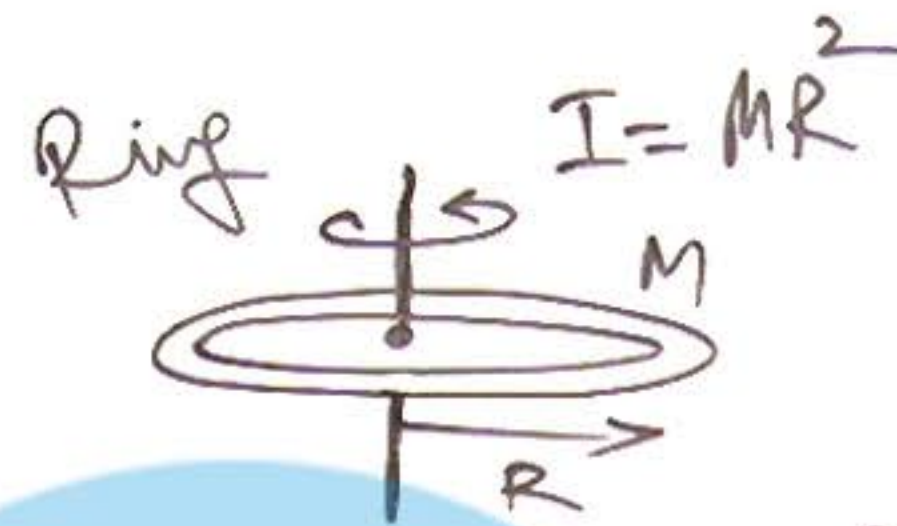
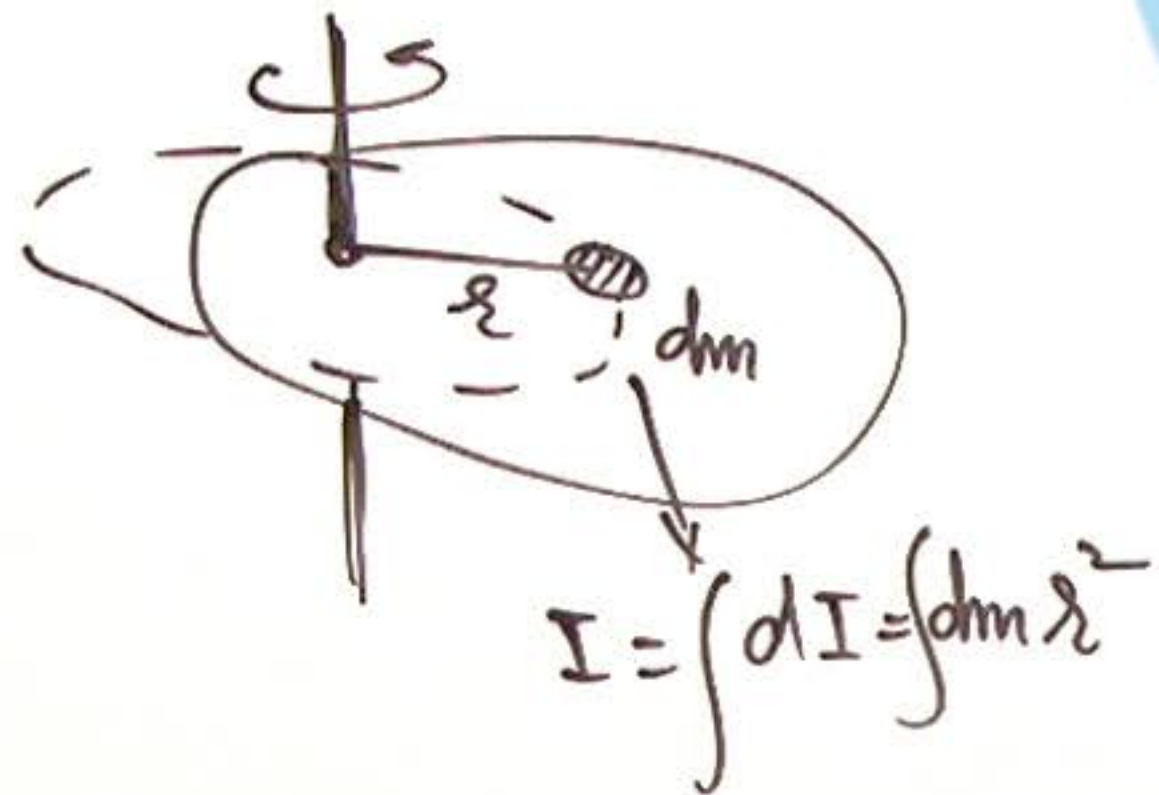
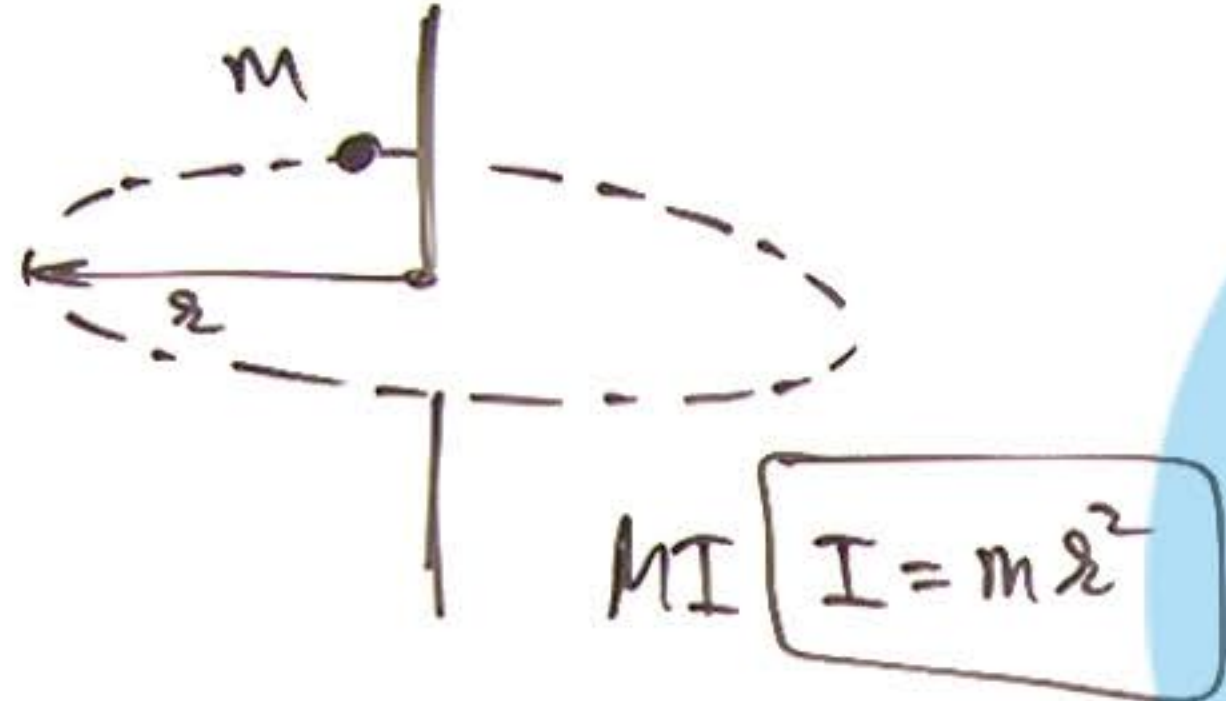
① Difference in Circular & → point mass

Rotational Motion:

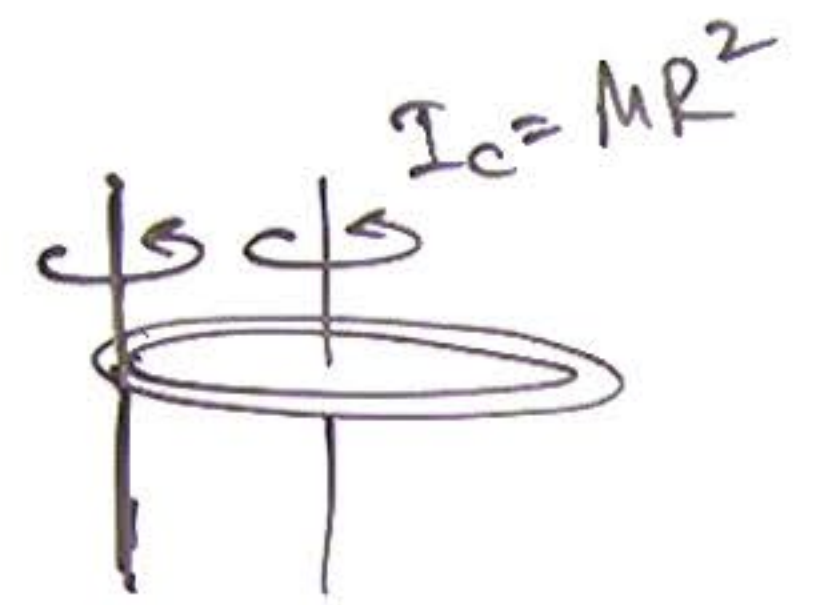
↓
extended
rigid body



2 Moment of Inertia:



③ Axes Theorems for MI:



$$I = MR^2 + MR^2 = 2MR^2$$



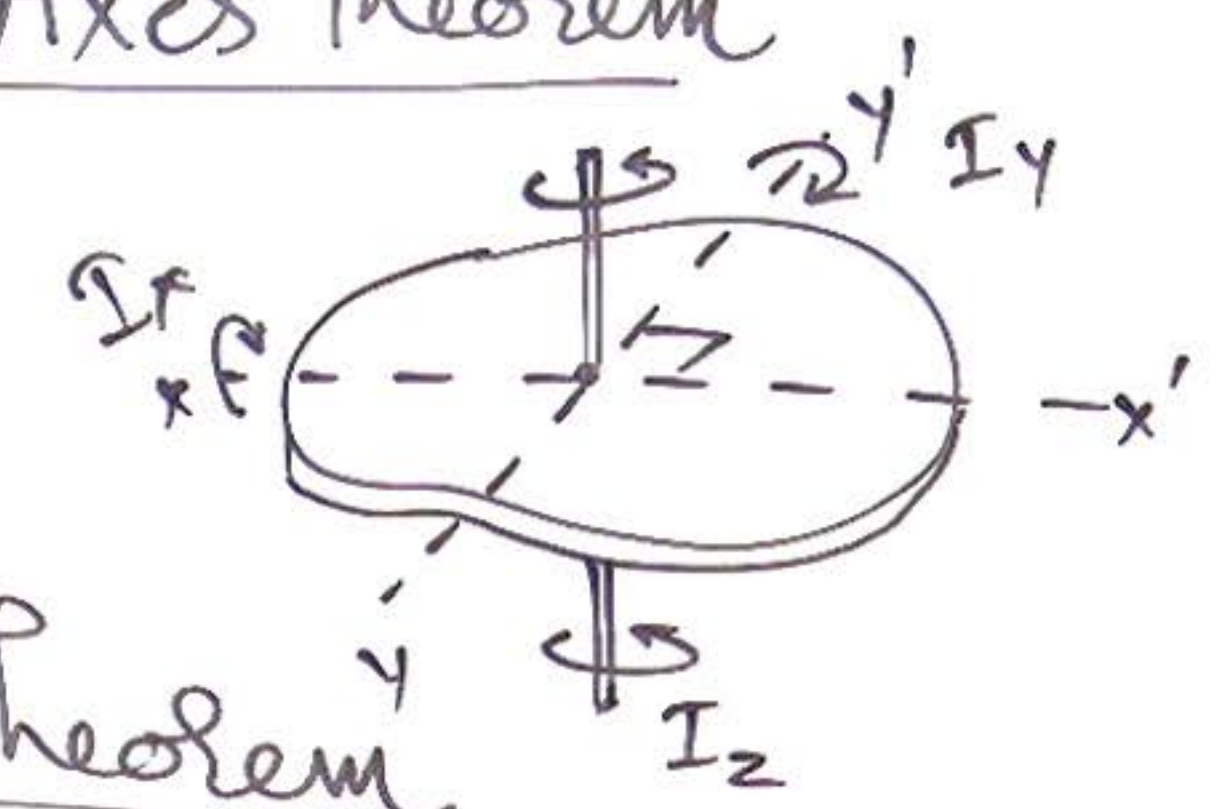
$$I = \frac{1}{2}MR^2$$



$$I = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$$

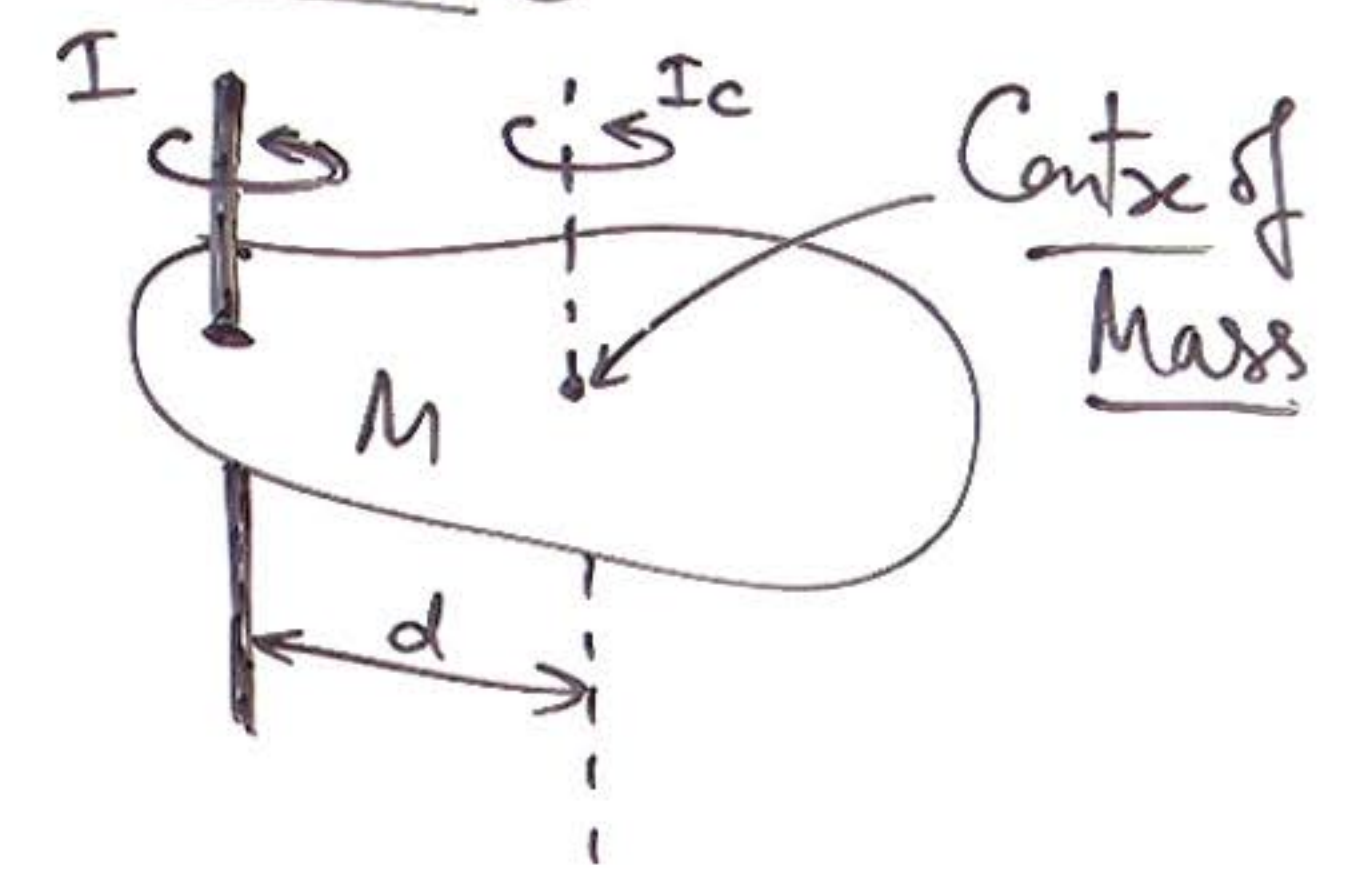
① Perpendicular Axes Theorem

$$I_z = I_x + I_y$$

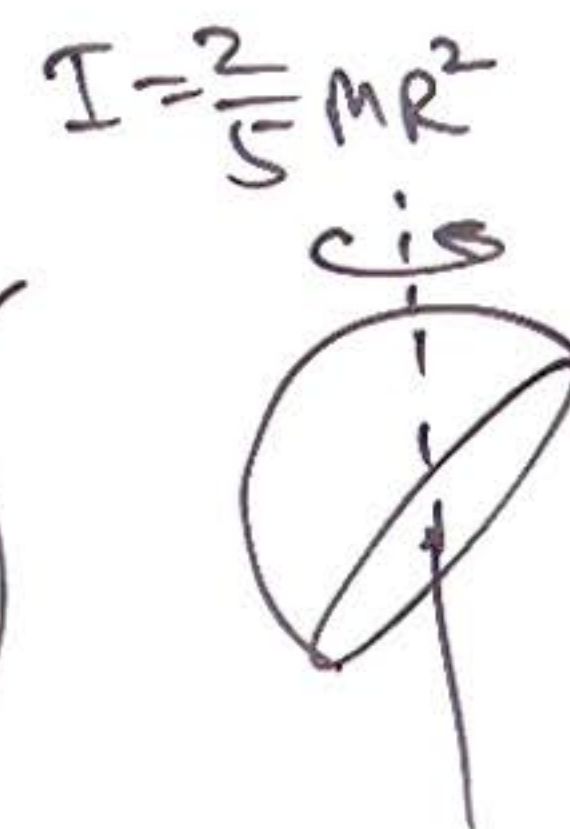
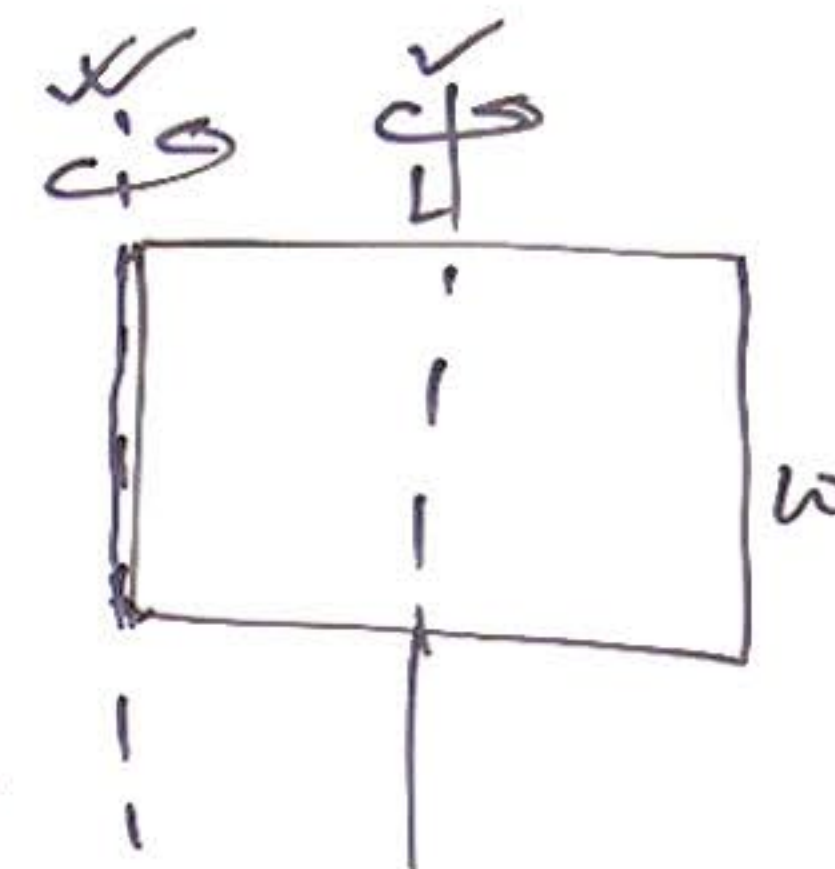
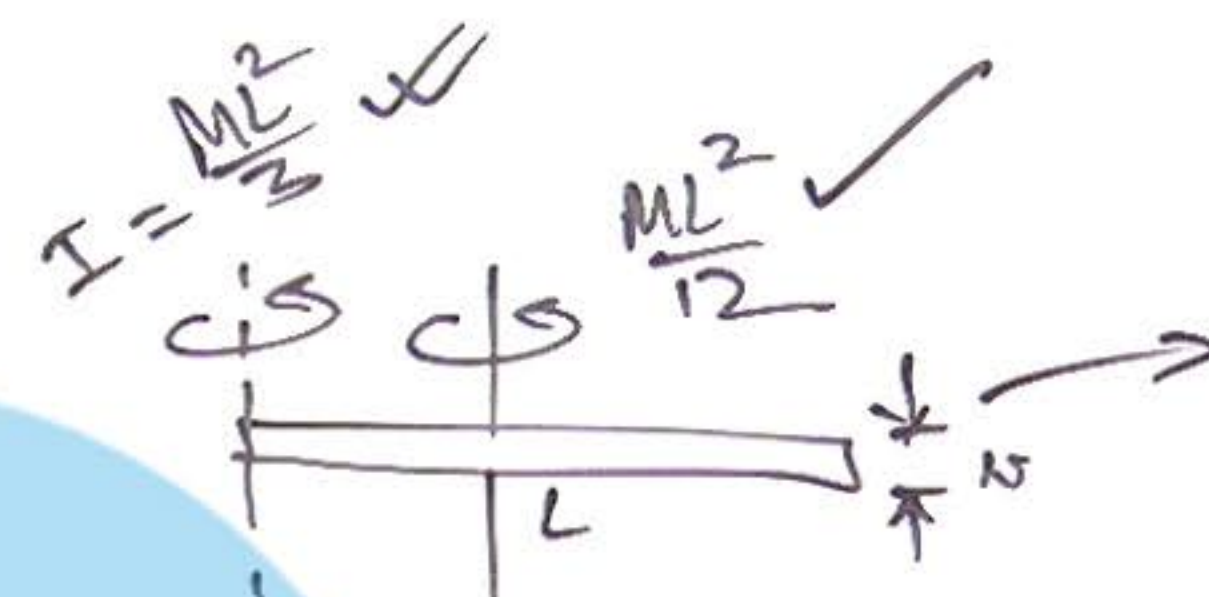
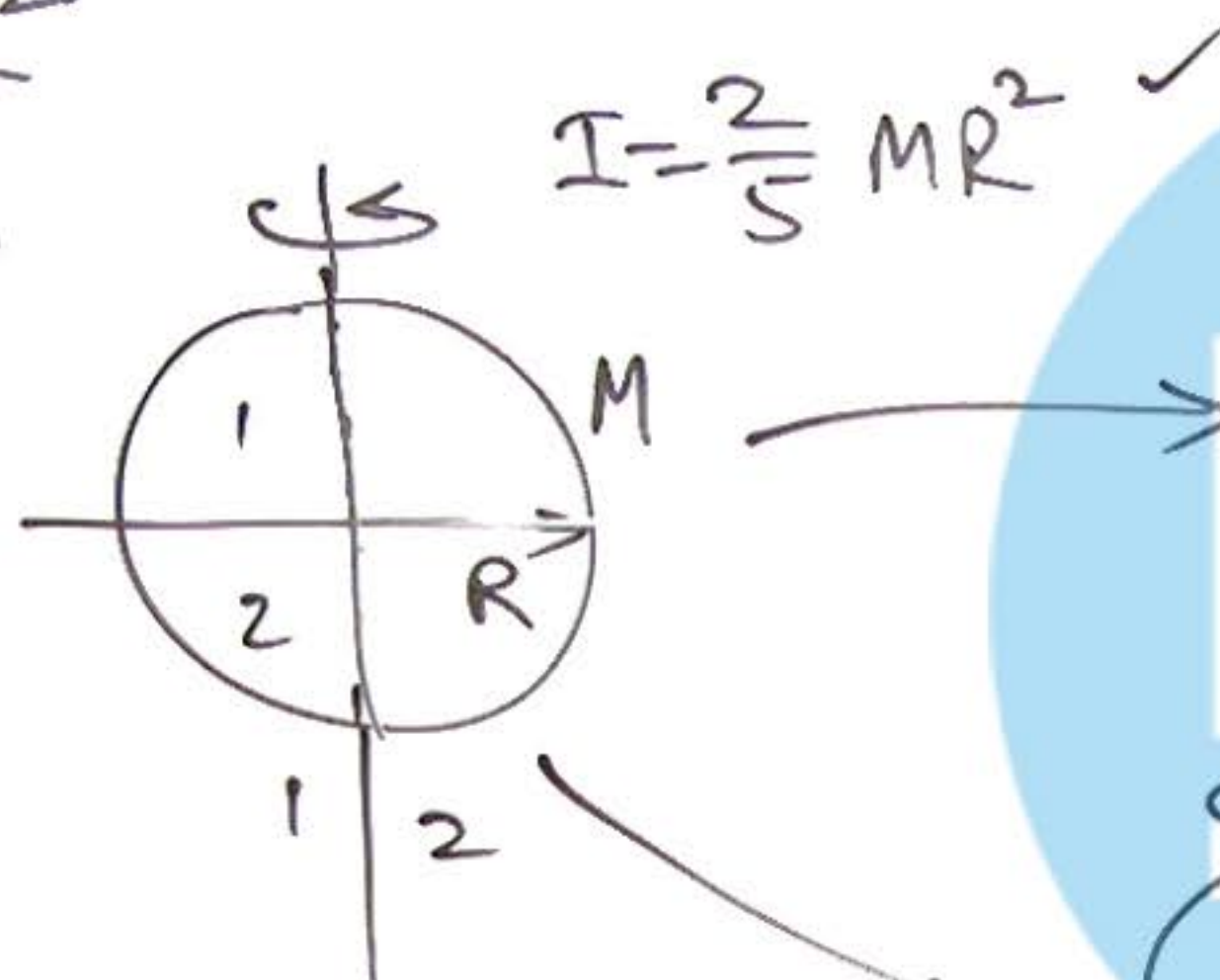
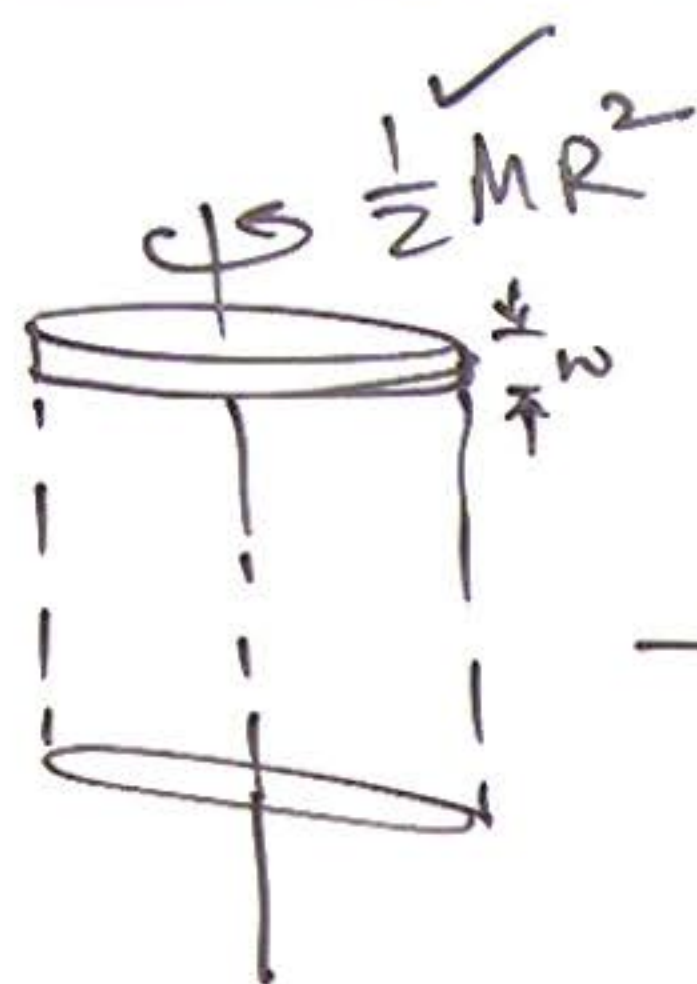


② Parallel Axes Theorem

$$I = I_c + Md^2$$



④ Mass Distribution property for MI:

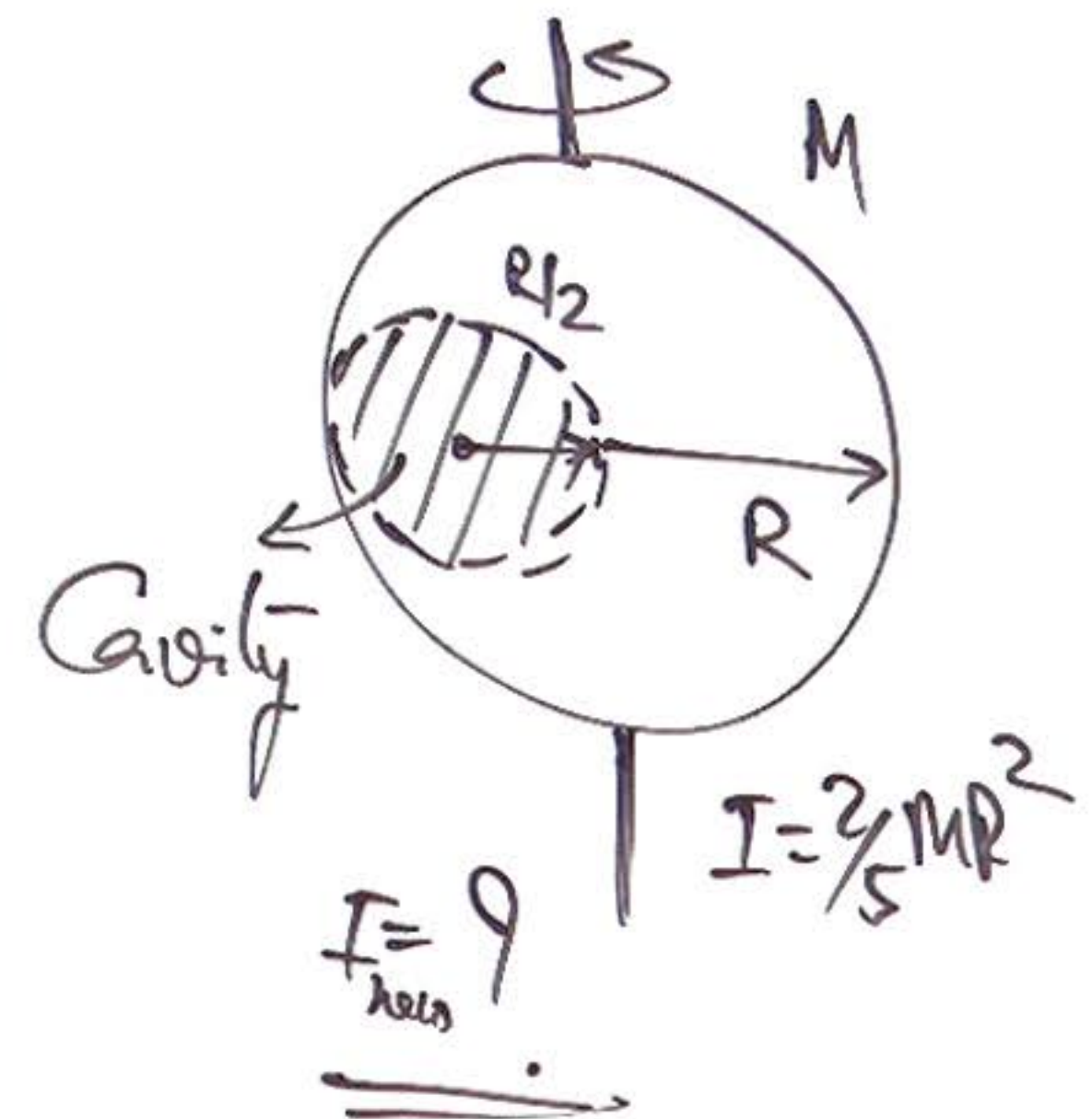
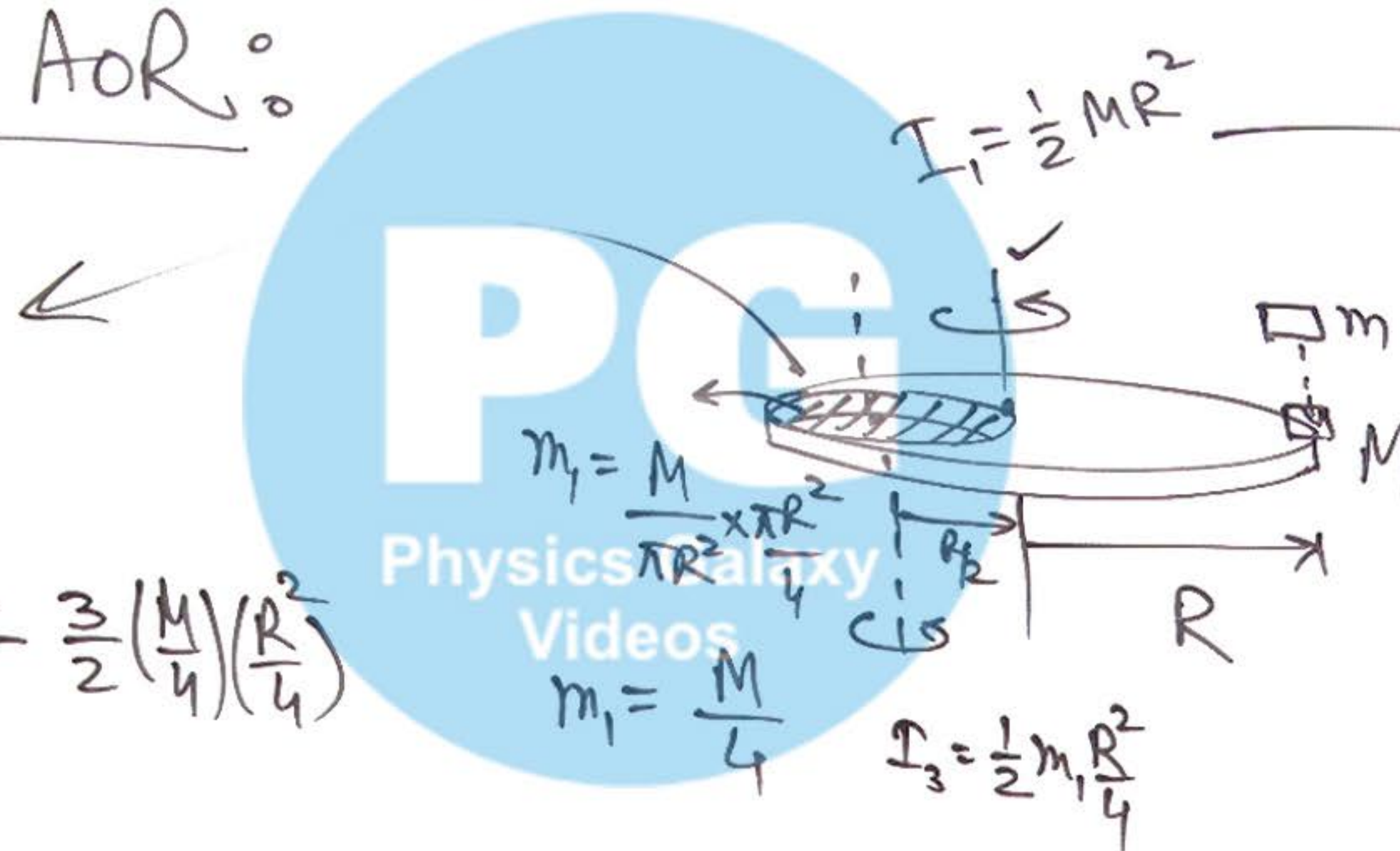


⑤ MI can be added or subtracted
about given AOR:

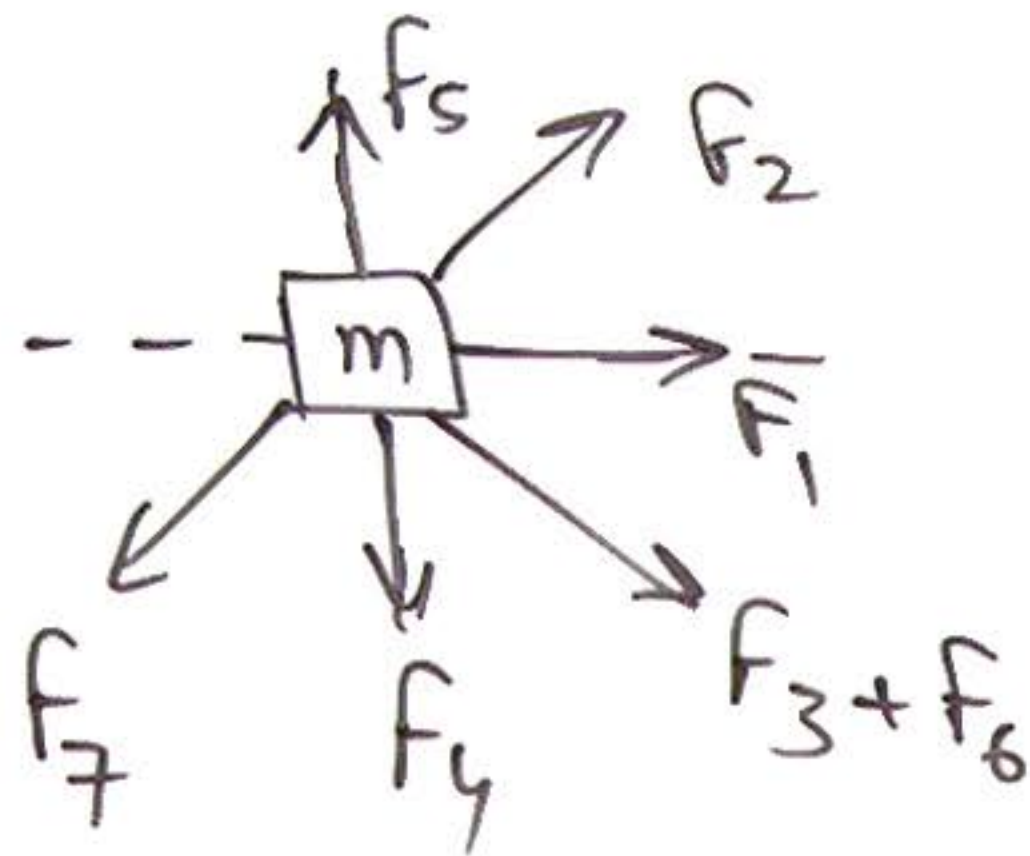
$$I_4 = I_2 - I_3$$

$$= \frac{1}{2}MR^2 + mR^2 - \frac{3}{2}\left(\frac{M}{4}\right)\left(\frac{R}{4}\right)^2$$

$$= \underline{\hspace{2cm}}$$

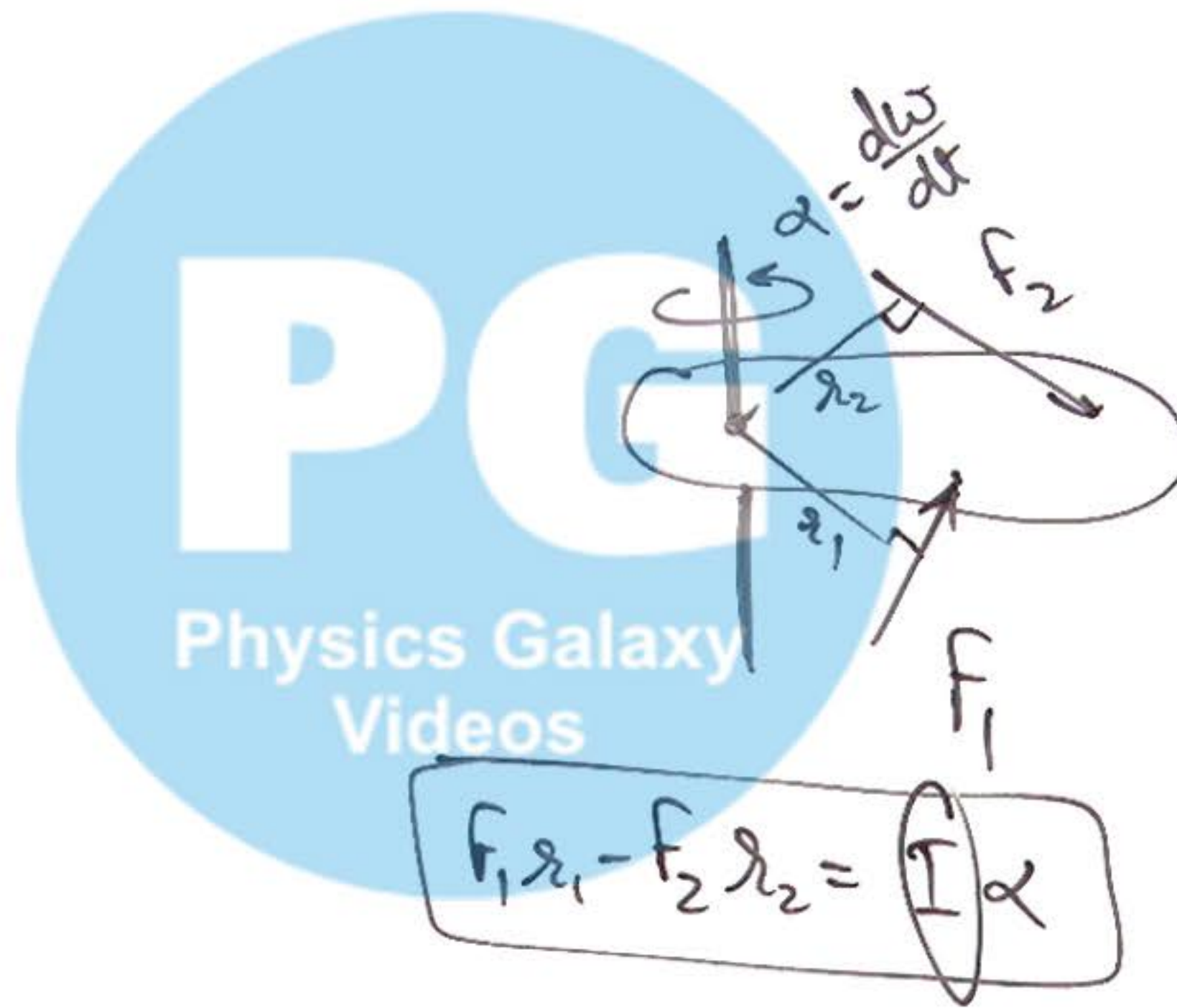


⑥ Newton's II Law in Rotational Motion:



$$\sum F_x = ma_x$$

$$\sum F_y = ma_y$$



$$F_1 r_1 - F_2 r_2 = I \alpha$$

NEL.

$$\sum \vec{\tau} = I \vec{\alpha}$$

⑥ Newton's II Law in Rotational Motion:

$$\boxed{a_t = R\alpha}$$

Diagram illustrating Newton's II Law in Rotational Motion for a cylinder and a block-and-string system.

Cylinder: A cylinder of mass m and radius R is suspended by a string. The moment of inertia is $I = \frac{1}{2}MR^2$. The forces acting on it are tension T (up) and weight mg (down). The acceleration is a (up) and the angular acceleration is α (clockwise). The relationship $a = R\alpha$ is noted.

Block and String System: A block of mass m_1 is on a smooth horizontal surface. A string is attached to it, passes over a pulley, and is attached to a hanging mass m_2 . The forces on m_1 are tension T (right) and weight m_1g (down). The acceleration is a_1 (right). The forces on m_2 are tension T (up) and weight m_2g (down). The acceleration is a_2 (down). The relationship $T \cdot R = I\alpha$ is noted.

Equations and Relationships:

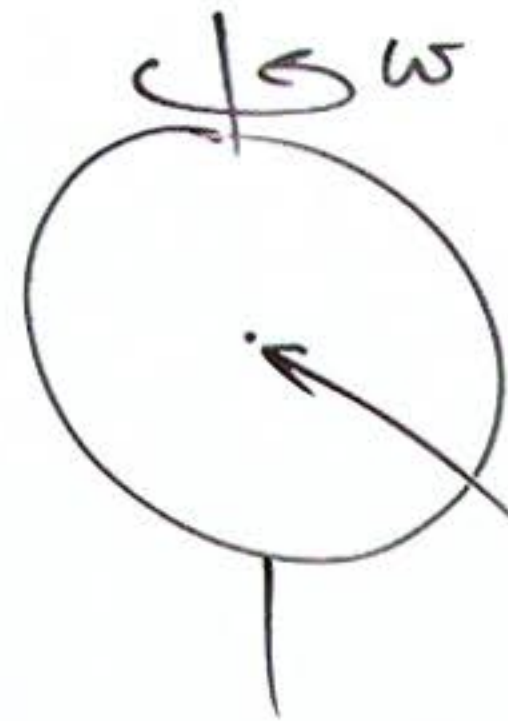
- For the cylinder: $mg - T = ma$
- For the block m_1 : $T = m_1a_1$
- For the hanging mass m_2 : $m_2g - T = m_2a_2$
- Rotational equation for the cylinder: $TR = (\frac{1}{2}MR^2)\alpha$
- Relationship between a and α : $a = R\alpha$
- Relationship between a_1 and a_2 : $a_2 = a_1 + R\alpha$
- Final acceleration for the cylinder: $a = \frac{2}{3}g$
- Final tension: $T = \frac{1}{2}MR \times \frac{g}{R}$
- Relationship between a and mg : $mg = \frac{3}{2}mg$

Boxed Equations:

$$\alpha = \frac{a_t}{R} = \frac{a_2 - a_1}{R}$$

⑦ Concept of "Static Equilibrium":

$$\left[\begin{array}{c} \vec{V}_{cm} = 0 \\ \& \\ \vec{\omega} = 0 \end{array} \right]$$



COM is at rest

$$\sum \vec{F}_{net} = 0$$

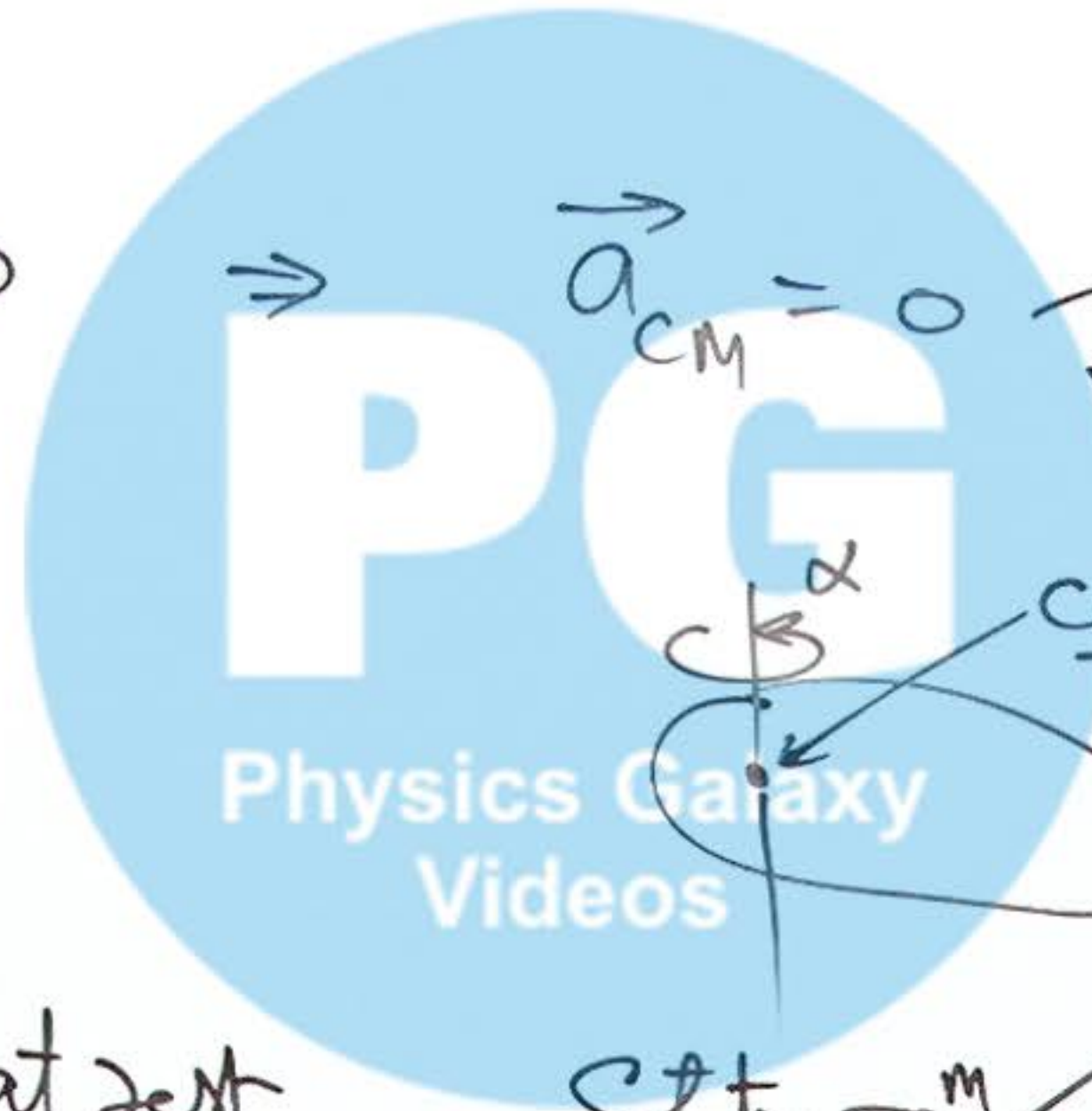
$\Rightarrow$

$$\vec{a}_{cm} = 0$$

$$\vec{V}_{cm} = 0$$

Static $\vec{e}_m$ of COM of body

$$\vec{V}_{cm} = \text{const}$$



Static $\vec{e}_m$

$$\sum \vec{F}_{net} = 0$$

$$\sum \vec{\tau}_{net} = 0$$

$$\left[\begin{array}{c} \vec{V}_{cm} = 0 \\ \& \\ \vec{\omega}_{body} = 0 \end{array} \right]$$